B.2 Invariance to Orthogonal transformation

Lemma B.4 (Invariance to Orthogonal transformation in the first coordinate)**.** *Let X and Y be two point clouds of the same size. Let U be an orthogonal transformation on the point space of X, X_U be the point cloud with each point in X transformed using U, then* $\mathsf{GNSA}(X, Y) = \mathsf{GNSA}(X_U, Y)$.

Proof. Let $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$, then $X_U = \{Ux_1, \ldots, Ux_N\}$. Let $\max_{x \in X}(d(x, 0)) = D_X$, $\max_{y \in Y}(d(y, 0)) = D_Y$ and $\max_{x \in X_U}(d(x, 0)) = D_{X_U}$. Since, each point in X_U is the U times each point in X. Then, we can write $D_{X_U} = \max_{x \in X}(d(Ux, 0))$. Since, for any two points, x_1 and x_2, $d(Ux_1, x_2) = d(x_1, U^T x_2)$, and $U^T 0 = 0$, then $D_{X_c} = \max_{x \in X}(d(x, 0))$. Hence, $D_{x_c} = D_X$. From the definition of GNSA,

$$\mathsf{GNSA}(X_U, Y) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \left(\frac{d(Ux_i, Ux_j)}{D_{X_c}} - \frac{d(y_i, y_j)}{D_Y} \right) \right|.$$

Again, using $d(Ux_i, Ux_j) = d(x_i, U^T U x_j)$, and $U^T U = I$,

$$\mathsf{GNSA}(X_U, Y) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \left(\frac{d(x_i, x_j)}{D_X} - \frac{d(y_i, y_j)}{D_Y} \right) \right|.$$

Hence, $\mathsf{GNSA}(X_U, Y) = \mathsf{GNSA}(X, Y)$. □

By Lemma A.2, we get invariance to orthogonal transformation in the second coordinate too and combining, we get the required lemma.

Lemma B.5 (Invariance to Orthogonal transformation)**.** *Let X and Y be two point clouds of the same size. Let U_1 be an orthogonal transformation on the point space of X, X_{U_1} be the point cloud with each point in X transformed using U_1. Similarly, let U_2 be an orthogonal transformation on the point space of Y, Y_{U_2} be the point cloud with each point in Y transformed using U_2, then* $\mathsf{GNSA}(X, Y) = \mathsf{GNSA}(X_{U_1}, Y_{U_2})$.

B.3 Not Invariant under Invertible Linear Transformation (ILT)

We can easily see this with a counter example. Let $X = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Let $Y = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$. Define A to be an invertible linear map such that $A(1, 0, 0) = (2, 0, 0)$, $A(0, 1, 0) = (0, 1, 0)$ and $A(0, 0, 1) = (0, 0, 1)$. Let X_A be the point cloud with each point in X transformed using A. Then from some calculation, we can see that $\mathsf{GNSA}(X, Y) = 0$ but $\mathsf{GNSA}(X_A, Y) = \frac{1}{9}$. Hence, we can see that GNSA is not invariant under Invertible Linear Transformations.

C Proofs for GNSA as a Loss Function

C.1 Differentiability of GNSA

To derive the sub-gradient $\frac{\partial \mathsf{GNSA}(X,Y)}{\partial x_i}$, we first define the following notation. Let $\mathbb{I} : \{T, F\} \to \{0, 1\}$ be a function that takes as input some condition and outputs 0 or 1 such that $\mathbb{I}(T) = 1$ and $\mathbb{I}(F) = 0$. Let $D_X = \max_{x \in X}(d(x, 0))$. Then it is easy to see that

$$\frac{\partial D_X}{\partial x_i} = \frac{x_i}{d(x_i, 0)} \mathbb{I}\{\arg\max_{x \in X}(d(x, 0)) = i\}.$$

Also, notice that $\frac{\partial d(x_i, x_j)}{x_i} = \frac{x_i - x_j}{d(x_i, x_j)}$, and that for $j \neq i$ and $k \neq i$, $\frac{\partial d(x_j, x_k)}{x_i} = 0$.

Let $m_X^{i,j} = \frac{d(x_i, x_j)}{D_X}$ and $m_Y^{i,j} = \frac{d(y_i, y_j)}{D_Y}$. Hence, combining, we get

$$\frac{\partial m_X^{j,k}}{\partial x_i} = \frac{\frac{\partial d(x_j, x_k)}{\partial x_i}}{D_X} - \frac{d(x_j, x_k) \frac{\partial D_X}{\partial x_i}}{(D_X)^2}, \text{ where these partial differentials are derived above.}$$

Lastly, notice that $\frac{\partial \left| m_X^{j,k} - m_Y^{j,k} \right|}{\partial m_X^{j,k}} = (-1)^{\mathbb{I}(m_Y^{j,k} > m_X^{j,k})}$.

Combining all the above, we get

$$\frac{\partial \mathsf{GNSA}(X,Y)}{\partial x_i} = \frac{1}{N^2} \sum_{1 \le j,k \le N} \frac{\partial \left| m_X^{j,k} - m_Y^{j,k} \right|}{\partial m_X^{j,k}} \frac{\partial m_X^{j,k}}{\partial x_i} \quad = \frac{1}{N^2} \sum_{1 \le j,k \le N} (-1)^{\mathbb{I}(m_Y^{j,k} > m_X^{j,k})} \frac{\partial m_X^{j,k}}{\partial x_i}.$$

Sub-gradients $\frac{\partial \mathsf{GNSA}(X,Y)}{\partial y_i}$ can be similarly derived.

C.2 Continuity

We prove the continuity of GNSA by showing that GNSA is a composition of continuous functions (Carothers, 2000, Theorem 5.10). Let

$$f_{i,j}(X,Y) = \left| \frac{d(x_i, x_j)}{\max_{x \in X}(d(x,0))} - \frac{d(y_i, y_j)}{\max_{y \in Y}(d(y,0))} \right|.$$

$$\text{It's easy to see that} \qquad \mathsf{GNSA}(X,Y) = \frac{1}{N^2} \sum_{1 \le i,j \le N} f_{i,j}(X,Y).$$

Since a sum of continuous functions is continuous, all we need to show is that $f_{i,j}(X,Y)$ is continuous for all $1 \le i, j \le N$. This is easy to see because $f_{i,j}(X,Y)$ can be seen as a composition of $d(\cdot,\cdot)$ and $\max(\cdot)$, along with the algebraic operations of subtraction, division and taking the absolute value. All the above operations are continuous everywhere[4], we get that their composition is also continuous everywhere. Hence, $\mathsf{GNSA}(\cdot,\cdot)$ is continuous.

C.3 Motivation for GNSA

The global part of NSA is inspired by Representational Similarity Matrix (RSM)-Based Measures (Klabunde et al., 2023). RSMs describe the similarity of the representation of each instance to all other instances within a given representation. The elements of the RSM $S_{i,j}$ can be defined using a valid distance or similarity function, formally: $S_{i,j} := s(R_i, R_j)$.

The motivation behind RSM-based measures is to capture the relational structure in the data. By comparing the pairwise similarities of instances within two different representations, RSM-based measures provide insights into how the internal structure of representations changes. RSMs are invariant to orthogonal transformations, which means they retain their structure under rotations and reflections. When using euclidean distance as a similarity function, the RSM is invariant under euclidean group $E(n)$, assuming translations.

One might think that the most intuitive approach to compare RSMs is to use a matrix norm. This method works well when the data is similarly scaled, as it is zero when the data is identical and larger when the data is dissimilar. However, it is not invariant to scaling, which can cause issues if the data is not similarly scaled. To ensure the data is on a similar scale, we first normalize the data using the normalization function:

$$R_{norm} = \frac{R}{2 * max(||x_i||)}$$

Where x_i is the point farthest from the origin in the space. One of the primary reasons for our metric's effectiveness is this normalization scheme. Although other normalizations are possible, NSA performs best when we normalize the space X by twice the distance between the origin of the space and the farthest point in the space. This ensures that all the points are within a unit sphere in N dimensions where N is the dimensionality of the space. The origin of the space is the point in $\mathbb{R}^N$ around which all points belonging to X are centered.

Most popular similarity measures, such as CKA(Kornblith et al., 2019) and RTD(Barannikov et al., 2022) also generate a similarity matrix. GNSA is simple in its implementation, but our choice of distance function and unique normalization scheme make GNSA a pseudometric, a similarity index and a valid, differentiable loss function.

[4]Note that division is not continuous when the denominator goes to zero. Since, the denominators are $\max_{x \in X}(d(x,0))$ and $\max_{y \in Y}(d(y,0))$, as long as neither of the representations is all zero, the denominator does not go to zero.

C.4 Improving Robustness of NSA

Formally, NSA is normalized by the point that is furthest away from the origin for the representation space. In practice, NSA yields more robust results and exhibits reduced susceptibility to outliers when distances are normalized relative to a quantile of the distances from the origin (e.g., 0.98 quantile). This quantile-based approach ensures that the distances are adjusted proportionally, taking into account the spread of values among the points with respect to their distances from the origin. It also ensures that any outliers do not affect the rescaling of the point cloud to the unit space.

D Proofs for LNSA as a premetric

Lemma D.1 (Identity)**.** *Let X be a point cloud over some space, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, X) = 0$.

Proof. Let $X = \{\vec{x}_1, \ldots, \vec{x}_N\}$. Hence, we see that

$$\mathsf{LNSA}^{\mathsf{metric}}(X, X) = \mathsf{LNSA}_X(X) + \mathsf{LNSA}_X(X) = 2\mathsf{LNSA}_X(X).$$

We show that $\mathsf{LNSA}_X(X) = 0$. We see that

$$\mathsf{LNSA}_X(X) = \frac{1}{N} \sum_{1 \leq i \leq N} \left(\mathsf{lid}(i, X)^{-1} - \overline{\mathsf{lid}}_X(i, X)^{-1}\right)^2.$$

Note that

$$\overline{\mathsf{lid}}_X(i, X) = \mathsf{AD}_X(i, kNN_X(i)) = \mathsf{lid}(i, X).$$

Hence we get

$$\mathsf{LNSA}_X(X) = \frac{1}{N} \sum_{1 \leq i \leq N} \left(\mathsf{lid}(i, X)^{-1} - \mathsf{lid}(i, X)^{-1}\right)^2 = 0.$$

And hence we have

$$\mathsf{LNSA}^{\mathsf{metric}}(X, X) = 2\mathsf{LNSA}_X(X) = 0.$$

□

Lemma D.2 (Symmetry)**.** *Let X, Y be two point clouds of the same size, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(Y, X)$.

Proof. Let $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$. Then

$$\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}_X(Y) + \mathsf{LNSA}_Y(X) = \mathsf{LNSA}^{\mathsf{metric}}(Y, X).$$

Hence,

$$\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(Y, X).$$

□

Lemma D.3 (Non-negativity)**.** *Let X, Y be two point clouds of the same size, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) \geq 0$.

Proof. Let $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$. Note that

$$\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}_X(Y) + \mathsf{LNSA}_Y(X),$$

and

$$\mathsf{LNSA}_X(Y) = \frac{1}{N} \sum_{1 \leq i \leq N} \left(\mathsf{lid}(i, X)^{-1} - \overline{\mathsf{lid}}_X(i, Y)^{-1}\right)^2.$$

Since, for all i, $\left(\mathsf{lid}(i, X)^{-1} - \overline{\mathsf{lid}}_X(i, Y)^{-1}\right)^2 \geq 0$, hence we get $\mathsf{LNSA}_X(Y) \geq 0$. Similarly, $\mathsf{LNSA}_Y(X) \geq 0$ and hence, $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) \geq 0$. □

From Lemma D.1, Lemma D.2 and Lemma D.3, we see that $\mathsf{LNSA}^{\mathsf{metric}}$ is a premetric over the space of point clouds.